

Final Report – Cryptocurrency Volatility Prediction

1. Introduction

Cryptocurrency markets are highly volatile and predicting volatility is important for risk management and trading strategies.

2. Problem statement

The objective of this project is to build a machine learning model that predicts cryptocurrency volatility using historical market data such as OHLC prices, trading volume and market capitalization.

3. Dataset

The dataset contains historical price and market information for multiple cryptocurrencies.

4. Data Preprocessing

Data cleaning included removing duplicates, handling missing values and removing irrelevant columns.

5.Exploratory Data Analysis

EDA was performed to understand price trends, correlations and volatility patterns.

6.Feature engineering

New features such as daily returns, rolling mean and volatility were created.

7. Model Training

A Random Forest Regression model was used to predict cryptocurrency volatility.

8.Evaluation

Model performance was evaluated using Mean Squared Error (MSE).

9.Conclusion

The model successfully predicts volatility patterns based on historical data.